A Scoping Review of Human–Computer Interaction Research on the Accessibility of Data Visualizations: New Technologies, Procedures, and Designs

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Abstract

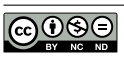
Data visualizations are vital for communication and decision-making, yet accessibility barriers limit their effectiveness for people with disabilities. While earlier literature reviews examined Human-Computer Interaction (HCI) research on visualization accessibility, there has not been an analysis that accounts for recent seismic shifts in software, hardware, design and experimental protocols due to Artificial Intelligence (AI), the COVID-19 pandemic, and greater awareness of accessible technologies. To analyze the impacts of this shift on accessible visualization research, we reviewed 280 HCI publications and coded target communities, visualization types, interaction types, and methods, contrasting articles published before and after 2021. Our corpus identified a sharp increase in publications after 2021 along with an expansion to a broader range of target communities, including people with CVD, IDD, and ADHD. Empirical and artifact contributions still dominate, with studies after 2021 showing a higher proportion of empirical studies. Participatory approaches are increasing but still remain limited. Research mainly focuses on charts and web-based, text-driven interactions, with articles after 2021 indicating a growing interest in multimodal and AI-driven systems. We further dissect this recent surge, positioning it against earlier work to identify trends, limitations, and open questions to guide future research.

CCS Concepts

• **Human-centered computing** → *Accessibility theory, concepts and paradigms*; *Visualization theory, concepts and paradigms.*

Keywords

Literature Review, Survey, Accessibility, Data Visualization



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1 Introduction

Data visualizations offer an efficient way to engage with quantitative information. They help people interpret insights about health, education, finance, news, and broader personal, community, and societal issues as data literacy expands [112, 152, 218]. By leveraging visual perception and preattentive processing, they communicate complex ideas and support decision-making more effectively than many other modalities [122].

Although visualizations are designed to make complex information more comprehensible, they often exclude large segments of the population, including people with disabilities [147]. People who are *blind or have low vision (BLV)* are among the most directly affected, as the visual modality is central to most visualization techniques [28]. People who are BLV experience varying levels of blindness, which shape their distinct accessibility needs for engaging with data visualization [85]. Accessibility barriers also extend beyond this community to people with *intellectual and developmental disabilities (IDD)* [270] as well as *cognitive disabilities* [178], who may face challenges when processing dense or abstract graphical representations. People who are *DeafBlind* often depend on tactile communication systems and devices as primary channels for accessing information [56]. People with *color vision deficiencies (CVD)* can struggle when designs rely solely on hue differences [15]. For people without disabilities, *situational impairments*, e.g., accessing visualizations on small mobile screens or in low-light conditions can also limit data visualization comprehension [211].

There is a growing push to make data visualizations more accessible, with contributions from industry and non-profit sectors

[112, 122]. Efforts range from implementing web accessibility standards [239] to developing commercial devices, such as dynamic tactile displays [114], and more recently, using deep learning and generative AI to create multimodal representations [124, 215, 283]. Within academic research, visualization accessibility has long been overlooked, particularly at premier venues where accessible design has lagged behind the field's rapid growth [147]. Still, the area is gaining traction as scholars in HCI, visualization, and related disciplines increasingly collaborate.

As work in this domain has increased, we are seeing more contributions that take the form of surveys, theoretical frameworks, guidelines, or opinion pieces [112, 157, 166, 239]. For example, researchers have employed the *crip technoscience* lens as a critical framework for reimagining how accessible data visualizations might be created [112], while other studies have examined how visualizations "in the wild" are made accessible (if at all) and the challenges practitioners face in designing them [122]. A handful of surveys have also reviewed accessibility in data visualizations, spanning both the HCI and visualization communities as well as underlying technologies explored primarily within the machine learning (ML) community. HCI and visualization surveys have outlined design spaces and identified open challenges [137, 248, 265], while ML-oriented surveys have focused on approaches such as alt text generation [35], automatic chart understanding [78], and data physicalization [64].

Over the past five years, research on accessible data visualizations has expanded rapidly in both volume and scope, with increasing contributions from HCI and accessibility communities, greater methodological diversity, and broader attention to disability groups beyond people who are BLV. There are many reasons why 2020-2021 may serve as an inflection point for accessibility research. The wide availability of natural language processing and image processing AI models has drastically lowered the barrier to entry for building accessibility software. The COVID-19 pandemic changed the ways of working for people across the globe, putting greater emphasis on technology for equal participation for all [253]. There has also been greater attention on novel methodologies for design and experimentation, including participatory design, design justice, and other inclusive design principles. We believe this is a timely moment to take stock of the field and examine how work published after 2021 differs from earlier efforts, particularly in terms of design and user engagement. To address this need, we present the most current survey of HCI research on accessible data visualizations, covering work through March 2026.

We present a scoping review and analysis of 280 publications on accessibility of data visualizations published between 1996 and March 2026. Our team, with expertise in accessibility, HCI, and visualization, reviewed research from the ACM Digital Library[1] , IEEE Xplore[2] , Eurographics proceedings in Wiley Online Library[3], Springer Nature Link[4], and Taylor and Francis Online[5]. We focused on HCI papers on accessible data visualization and used descriptive and qualitative analyses to identify trends before and after 2021.

[1]ACM Digital Library: https://dl.acm.org/
[2]IEEE Xplore: https://ieeexplore.ieee.org/Xplore/home.jsp
[3]Wiley Online Library: https://www.wiley.com/en-us
[4]Springer Nature Link: https://link.springer.com/
[5]Taylor and Francis Online: https://www.tandfonline.com/

Our review was guided by four questions that build on prior surveys [137, 265] and help identify broader thematic changes in this research in the past five years.

RQ1: How has the consideration of and engagement with *participants and target populations* in HCI research on accessible data visualization within computing venues changed over the last five years?

RQ2: What *technologies* involved in accessible visualizations have changed most drastically in the last five years?

RQ3: How has the volume of published HCI research on accessible data visualization within computing research communities changed in recent years?

RQ4: What open questions and future research opportunities emerge from this review?

To investigate these research questions, we coded 280 papers gathered in our corpus, as described in Section 3. The contributions of this work lie in both the scoping review of these 280 publications and the qualitative analysis of current trends and open questions that can guide future research. We conduct a quantitative analysis of the resulting code frequencies, comparing prior to 2021 and 2021 & beyond in Section 4. We then present qualitative findings derived from close readings of the corpus, highlighting thematic changes in research on the accessibility of visualizations in Section 5. Finally, we discuss open questions and identify future research directions in Section 6.

2 Background and Related Work

2.1 Literature Reviews in HCI, Accessibility, and Visualization Research

Literature reviews and surveys have long been important contributions in HCI [237], accessibility [159], and visualization [153, 170, 197]. They synthesize prior work, surface trends, and articulate shared conceptual frameworks [266]. We situate our survey within this tradition to provide methodological grounding and context.

Reviews in *HCI* often analyze large corpora of publications to trace research trajectories and shifts in the field (e.g., [37, 154]). Most use qualitative coding to derive themes. Common dimensions include contribution types, user communities, and research methods. Reviews also extract information such as topical focus [58] and participant sample sizes [36]. Some incorporate quantitative techniques, including clustering and network analysis, to map trends [154], and others examine alternative corpora such as patents [37]. Despite these variations, qualitative synthesis remains the dominant approach in HCI.

A similar pattern appears in *accessibility* research. Reviews in this area often combine qualitative coding with descriptive statistics to report on communities of focus, research problems, contributions, participant engagement, researcher roles, and temporal trends [159]. Large-scale surveys consistently show that work on accessibility for people who are BLV receives the most attention, followed by research on motor impairments and Deaf or Hard-of-Hearing (DHH) communities [210]. Smaller reviews report similar findings [14]. Mack et al. [159] also note that usability studies, interviews, and controlled experiments dominate, while participatory methods remain comparatively rare.

Surveys in *visualizations*, such as State-of-the-Art Reports (STARs), similarly examine research trajectories, often in emerging areas, by employing qualitative coding of survey corpora [137] to derive common themes and patterns of design. Specifically, surveys on data visualizations often surface characteristics of design [209], taxonomies [82], and guidelines and principles for future design [137].

Our review sits at the intersection of these three areas, building on literature survey practices commonly employed in HCI research. We focus on ❶ HCI research that examines ❷ data visualization in the context of ❸ accessibility for people with disabilities.

2.2 Relevant Surveys, Theoretical Perspectives, and Opinion Papers

We summarize the insights from related surveys presented in Table 1. We identify Wimer et al. [265], Kim et al. [137], and Tlapa-Monge and Montané-Jiménez [248] as directly comparable to our study, as they are the closest survey works at the intersection of accessibility, HCI, and visualization. Other relevant surveys are highlighted later in this section but were not directly comparable to ours due to differences in scope, including a focus on non-visualization domains, modality-specific systems, or machine learning tasks.

Most closely related, Kim et al. [137] mapped the design space of accessible visualization for BLV readers across seven dimensions and proposed a four-part model for accessibility. Similarly, Tlapa-Monge and Montané-Jiménez [248] evaluate the space of existing solutions for BLV accessibility, synthesizing practical implications and design guidelines. Wimer et al. [265] take a broader view of accessibility to include non-visual disabilities, creating a corpus of research for people who are BLV as well as people with non-visual disabilities. Using the full corpus, they present a classification scheme for accessible data representations and then subsequently reduce their corpus to articles focused on communities with non-visual disabilities for an in-depth synthesis of the current state of research. Together these three surveys provide a comprehensive prior synthesis of the space of accessible visualization research across disability communities.

A cross-corpus comparison with these three surveys reveals limited overlap with our corpus (11.4% with Kim et al. [137], 15.6% with Wimer et al. [265], 8.2% with Tlapa-Monge and Montané-Jiménez [248] and 27.9% across all three)[6], despite their conceptual proximity. This reflects differences in scope and temporal coverage—all reviews conclude prior to 2023, with the most similar ([137]) concluding prior to 2021—whereas our survey extends through 2026 with HCI-focused inclusion criteria. Together, this highlights the rapid expansion of the field and positions our work as a comprehensive and up-to-date synthesis of accessibility research in data visualization.

The primary way we extend prior research is by adopting a scoping review approach common in recent HCI and accessibility field or domain–spanning surveys (e.g., [10, 159]) to examine accessibility within the data visualization literature. While prior surveys have primarily developed taxonomies of techniques and models, our approach focuses on engagement with design considerations, user engagement, and study details. We explicitly compare publications pre- and post-2021, a bifurcation motivated in part by notable changes in the landscape of research beginning around 2021, from factors such as the availability of AI and the COVID-19 pandemic, as well as by the temporal scope of the directly comparable surveys, to assess how earlier agendas have materialized and to outline open HCI research questions.

[6]The underlying data for these calculations is provided in the supplementary material.

2.2.1 Surveys of Research on Automatic Chart Understanding and Alt Text Methods. ML researchers have surveyed automatic tools for analyzing and converting visualizations, outlining pipelines for chart extraction, classification, data extraction, and higher-level interpretation [57]. These reviews categorize work by chart types, data extraction methods, modality-based techniques, traditional image processing, and deep learning approaches [78, 217]. They also highlight applications such as chart redesign, captioning, and chart question answering, along with persistent challenges. Complementary surveys examine technologies for making figures accessible, such as alt text generation tools [276]. Together, insights from these surveys show that while many tools exist for readers and creators, performance gaps remain and must be addressed for reliable use, and they illuminate complexity dimensions that inform HCI work relying on off-the-shelf ML solutions.

2.2.2 Surveys of Research on Haptics, Tactile Displays & Data Physicalization. Researchers have also surveyed a range of haptic, tactile, and multimodal approaches for making data visualizations accessible [161]. O'Modhrain et al. [183] studied existing literature to identify the challenges and opportunities with designing visual media for refreshable touch displays. Early reviews of haptic visualization [184, 205] traced historical milestones (e.g., Project GROPE, PHANTOM), advances in haptic rendering and perception, and applications across visualization, while more recent surveys [119] highlight growing use of haptic technologies along with challenges related to efficiency, detail extraction, memory load, and support for complex visualizations. Other reviews examine touch-based graphics in STEM contexts [34]. Surveys of data physicalization [20] further expand beyond screen-based tools by showing how form, volume, texture, vibrotactile signals, and light can supplement encodings. Broadly speaking, these reviews highlight multimodality as central to advancing accessibility in data visualization.

2.2.3 Relevant Surveys and Perspectives on Disability-Focused Data Visualization Research. Researchers have argued for expanding the intellectual scope of accessibility and visualization research by moving beyond narrow paper types and recognizing a broader set of twenty-five contribution types, including interaction techniques, new domains, and visions, to better capture how visualization advances knowledge [148].

Torres and Barwaldt [250] systematically reviewed approaches for making diagrammatic representations accessible to BLV users and found that most work centered on auditory interfaces and refreshable braille displays, with limited attention to navigation, editing, or collaboration. Complementing this, Vargas and Trevisan [254] surveyed 82 studies on making graphics, diagrams, flowcharts, shapes, maps, and educational images accessible for BLV users, categorizing work by task, modality, and assistive technology, and identifying ongoing challenges with automation, real-world evaluation, and the need for more personalized multimodal solutions. While both of these surveys aim to support visual accessibility,

Table 1: Comparison of prior relevant literature surveys. The first group of rows contains the most relevant and directly comparable surveys, with subsequent rows grouped by research communities (HCI & Visualization, Machine Learning, and Haptics and Data Physicalization). The Review column identifies the author(s) and publication year. Venue indicates the publication venue. Papers reports the number of works included in each review. End Year reports the final year included in the survey corpus. Comm. refers to the communities considered, categorized as *BLV* (Blind or Low Vision), *DLD* (Developmental Language Disabilities), *Broad* (multiple disability communities or general populations), or *None*. Vis. Types are simplified into *Charts* (a single type) or *Multiple* (two or more types). Auth. captures whether authoring practices or tools for creating accessible visualizations were discussed. AI indicates whether automation or machine-learning-based approaches were reviewed. Finally, User Studies denotes whether findings from empirical studies were synthesized. All checkmarks, along with entries coded as *Broad* and *Multiple*, are highlighted to mark coverage. Three works at the intersection of HCI, visualization, and accessibility are identified as direct comparables, while four others at this intersection (identified by #) are not comparable because of differences in scope.

Review	Venue	Papers	End Year	Comm.	Vis. Types	Auth.	AI	User Studies
Direct Comparables								
Hassan et al. (This review)	ASSETS	280	2026	Broad	Multiple	✓	✓	✓
Wimer et al. [265] (2024)	TVCG	152	2022	Broad	Multiple	✓	✓	✓
Kim et al. [137] (2021)	EuroVis	56	2020	BLV	Multiple	✗	✗	✓
Tlapa-Monge & Montané-Jiménez [248] (2023)	CONISOFT	45	2022	BLV	Multiple	✓	✓	✓
Visualization & HCI—Not Comparable[#]								
Enge et al. [70] (2024)	EuroVis	57	2023	None	Multiple	✗	✗	✓
Vargas and Trevisan [254] (2024)	IHC	82	2023	BLV	Multiple	✗	✗	✓
Devane et al. [59] (2024)	IJLCD	6	2023	DLD	Multiple	✗	✗	✓
Torres and Barwaldt [250] (2019)	FIE	26	2018	BLV	Diagrams	✓	✗	✓
ML Chart Understanding & Alt-Text Generation								
Yan et al. [276] (2025)	IEEE Access	38	2024	BLV	Charts	✓	✓	✓
Farahani et al. [78] (2023)	IEEE Access	~120	2023	None	Charts	✗	✓	✗
Shahira and Lijiya [217] (2021)	IEEE Access	–	2020	BLV	Charts	✓	✓	✗
Davila et al. [57] (2020)	TPAMI	~130	2019	None	Charts	✗	✓	✗
Haptics, Tactile Displays, & Data Physicalization								
Jiang et al. [119] (2025)	TACCESS	132	2024	BLV	Multiple	✓	✗	✓
Mallouk et al. [161] (2024)	ICICT	–	2024	None	Multiple	✗	✗	✗
Butler et al. [34] (2021)	CHI	292	2020	BLV	Multiple	✓	✗	✓
O'Modhrain et al. [183] (2015)	ToH	–	2015	BLV	Multiple	✓	✗	✗
Bae et al. [20] (2022)	CHI	47	2021	Broad	Multiple	✓	✗	✓
Paneels and Roberts [184] (2009)	ToH	~70–100	2009	BLV	Multiple	✗	✓	✗
Roberts and Paneels [205] (2007)	WHC	–	2006	BLV	Multiple	✓	✗	✗

they differ from ours in scope as they do not focus directly on data visualizations.

3 Method

Since the goal of this work was to understand research trends, we employed a scoping review as our research method [19]. Like systematic reviews, a scoping review involves analyzing how research is conducted, summarizing findings, identifying gaps, and presenting open questions. However, unlike a systematic review, it does not assess the quality of the studies, as its goal is to capture the full scope of the literature, including artifacts [177, 188]. We also adapt elements of systematic literature review checklists, including PRISMA, that are identified as relevant for scoping reviews by prior work [252]. We describe our literature search, screening process, inclusion and exclusion criteria, data extraction, and coding in detail in this section.

3.1 Selection of Digital Libraries

Our selection of digital libraries was informed by prior surveys we build upon [137, 265] as well as our team's expertise, which is largely grounded in visualization, accessibility, and HCI research. We began by focusing on the digital libraries emphasized in prior work [137, 265], which are also primary outlets for accessibility-focused visualization and HCI research: ACM and IEEE. Specifically, we included the full ACM Digital Library, which hosts major HCI and accessibility venues such as ASSETS, W4A, and CHI (including accessibility- and visualization-related tracks and committees). IEEE Xplore was included for its leading visualization venues, including TVCG and VIS, as well as related conferences central to the field. Following an initial pass, we expanded our scope by reviewing citations and references from papers in these core libraries and through internal consultation. This led to the inclusion of Eurographics (via Wiley Online Library), given its longstanding role

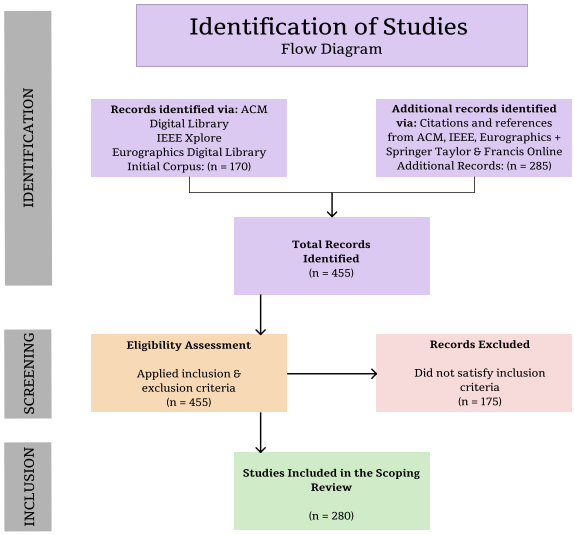


Figure 1: An overview of the corpus identification and screening process using a PRISMA-style flow diagram. Initially, we performed targeted keyword searches in the ACM Digital Library, IEEE Xplore, and Eurographics Digital Library, yielding an initial corpus of 170 records. We then expanded the corpus by searching the citations and references of the initial records and conducting additional targeted searches in Springer Nature Link and Taylor & Francis Online, yielding 285 additional records and 455 records in total. Next, we assessed all records against our inclusion and exclusion criteria for HCI contributions toward data visualization accessibility. We excluded 175 records that did not meet the inclusion criteria, yielding a final corpus of 280 studies included in the scoping review.

in graphics and visualization and its recurring contributions to accessibility research. We also added Springer Nature Link, which hosts several relevant HCI venues such as Universal Access in HCI and HCI International, as well as Taylor & Francis, which publishes the International Journal of Human–Computer Interaction.

Our team does not include experts in disability studies, so we did not examine disability-focused journals and venues. We also constrained our corpus to maintain a consistent coding rubric, particularly for descriptive HCI research methods and contributions [266] and for interaction-modality taxonomies described in prior work. We acknowledge that this choice limits cross-disciplinary insights but enables a focused view of recent trends and recent developments within HCI research on accessible data visualization.

In the following, we describe our dataset collection, curation, and coding framework.

3.2 Dataset Collection and Curation

Two of the authors met to systematically construct a list of keywords based on their domain expertise and prior taxonomies of accessibility in visualization literature [78, 137, 250].

We constructed three types of keywords: general terms such as Data Visualization* AND Disabilit* and Information Visualization* AND Accessibility; disability-specific terms using Visualization* AND paired with Blind*, Low Vision, Impairment*, Color Vision Deficienc*, Color Blind*, Dyslexia, Aphasia, ADHD, Autism, Deaf, Hard of Hearing, Deafblind*, Intellectual Disabilit*, Cognitive Disabilit*, and Mobility AND Limited; and method-focused terms such as Screen Reader, Tactile, Sonification, Physicalization, and Captions combined with Data Visualization*, as well as Charts, Plots, and Graphic combined with Accessibility. We used the "Advanced Search" feature in libraries to run these queries in abstracts and titles only, and manually reviewed the abstract of each result to assess relevance. Minor adjustments were made to the search terms to fit the configurations of each digital library.

We began our search with ACM, IEEE Xplore, and Eurographics, yielding 170 papers after screening. To improve coverage given the diverse terminology at the intersection of accessibility and visualization, we expanded the corpus by examining both the reference lists of these papers and the works citing them, restricting inclusion to venues within ACM, IEEE, or Eurographics.

To support this process, we developed three scripts: one to convert BibTeX entries into a Python-readable tabular format, and two using Selenium to automatically retrieve references and citations from ACM and IEEE. For Eurographics, parts of this process were completed manually.

During this expansion, we identified additional HCI venues outside our initial scope, including HCI International (HCII) and Universal Access in Human-Computer Interaction (UAHCI) (Springer Nature Link), and the International Journal of Human–Computer Interaction (IJHCI) (Taylor & Francis). We therefore included these digital libraries and applied the same keyword-based search strategy. This process resulted in an extended corpus of 455 papers, of which 280 were retained after applying the exclusion criteria described in the next section. Figure 1 summarizes the identification, screening, and inclusion process.

3.3 Paper Screening, Selection, and Eligibility Criteria

The initial screening was based on titles and abstracts, followed by a full-text review. When needed, two authors collaboratively discussed a subset of papers to reach agreement.

Our inclusion criteria were:

IC1 We include HCI work that makes at least one of the seven recognized contribution types in HCI [266]. Accordingly, our corpus includes not only publications reporting empirical findings but also those that contribute artifacts, including, digital visualization prototypes, systems, and designs spanning web, mobile, and desktop platforms; tactile graphics; 3D-printed or fabricated models; sonified or audio-haptic representations; or multimodal surfaces[7]. We additionally

[7]We focus on artifacts following the ACM SIGCHI definition of HCI as a discipline concerned with the design, evaluation, and implementation of interactive computing systems for human use, along with the study of the phenomena surrounding them. Artifact contributions in HCI present new inventions, e.g., systems, techniques, or designs, that enable novel opportunities or insights. They are typically instantiated as

include contributions that introduce datasets and benchmarks[8].

IC2 We include research on data visualization (graphical representation of data and information through visual elements, e.g., charts, maps, and dashboards).

IC3 The work must focus on people with disabilities or disability communities (e.g., content creators designing for people with disabilities).

During the final screening stage, we excluded papers for several reasons.

EC1 We removed work that focused on "visualization for accessibility" rather than the accessibility of data visualizations, e.g., systems that use visual markers to indicate a speaker in a virtual environment [151] or techniques supporting wayfinding and navigation [179]; in these cases, the visual content conveys contextual, spatial, or conversational cues rather than structured data intended for analysis.

EC2 We excluded work that did not include support for data visualization, including schematics [88], structural diagrams [185], or visual cues that do not encode data [186], because their accessibility requirements differ from those of data visualizations.

EC3 We removed papers that mentioned accessibility only as future work. We included work where improving the accessibility of data visualizations was an explicit goal, even without evaluation. However, we excluded papers that referenced accessibility merely as a possible application, such as chart-summarization models lacking empirical grounding [141], because including such work would dilute our analysis.

EC4 We excluded papers whose contributions consisted solely of computing algorithms, even when intended for data visualization (e.g., pipelines for chart classification and data extraction [4]), as algorithm-only work fell outside our scope of artifacts within HCI.

3.4 Coding Framework and Process

We analyzed our dataset using a combination of (1) descriptive summarization and (2) individual publication reviews as recommended by methodological work on scoping reviews [19]. We developed a coding rubric through an iterative process conducted across two meetings. Building on prior work [10, 159] and domain-specific considerations [137, 265], we designed our rubric to include the following categories:

(1) **Publication Metadata**, including title, year, venue, page length, DOI, and length of paper, coded as either full or short.
(2) **Types of Contributions**, recorded across three columns: primary, secondary, and other stated contributions. We adopted the contribution categories from [266], including empirical, artifact, methodological, theoretical, benchmark/dataset, survey, and opinion.
(3) **Communities of Focus**, captured in four columns: final-users-engaged (whether ultimate users were engaged, coded as Yes or No), primary, secondary, and additional communities.
(4) **Visualization-Specific Categories** captured visualization type, interaction modality, interface design, and AI use. After the initial coding, we consolidated codes into broader groups for insights, which were in turn defined directly from the coded content. Visualization types included charts, maps, tables, diagrams, broad focus, and an "other" category preserving original labels. Interaction modalities included tactile, haptic, refreshable tactile displays, and braille (later merged for some analyses); text-to-speech/screen readers and question–answer systems (also merged); and sonification. We added an additional code to indicate when a paper used multiple modalities. For interface and design, secondary codes covered surfaces, e.g., web, mobile, and tablets; head-mounted displays and wearables; braille and 3D prints; and applications, e.g., notebooks and Excel.
(5) **Research Methods** were recorded across three separate columns: primary, secondary, and other methods (for cases with more than two), as well as evaluation methods. Since papers conceptualized methods in diverse ways, we conducted a second round of categorization. We collapsed methods into broader methodological groups by merging highly similar forms, e.g., grouping all interview variants (semi-structured, guided, informal); combining all user studies that did not use a controlled experimental design or another specific qualitative approach, e.g., task-based and observational studies; and consolidating all literature survey methods (literature analyses, systematic reviews), while retaining distinctions for participatory design, co-design[9], automatic analysis, corpus collection, and longitudinal work.
(6) **Participant Information**, including how participants were engaged, the number of participants (with breakdowns across groups when applicable), the recruitment source (universities, organizations, schools, mailing lists), and whether demographic information or visualization expertise was reported.
(7) **Contribution Type**, describing the nature of the contribution and the main findings.

The two authors independently coded an initial subset of (N=16, 5.71 %) papers using the draft rubric motivated by prior work [10, 137, 159, 168], discussing both the categorizations and their interpretations to calibrate their approach. While no changes were made to the column names, the authors refined their coding strategies to ensure consistency, particularly in how scope and definitions were applied. Although prior reviews offered useful starting points, our rubric differed by combining HCI contribution types with methodological detail and visualization taxonomies for modalities and visualization types, whereas earlier work typically examined only one of these dimensions. These discussions helped refine how we categorized contributions (e.g., determining when

prototypes, mockups, demos, or other envisionments of a previously unseen idea, and may be partially or fully functional.

[8]Following Wobbrock and Kientz [266], benchmark or dataset contributions provide new corpora—often with accompanying analysis—to support future research. Benchmarks are typically paired with standard tests that enable comparison across projects, while datasets facilitate the evaluation of algorithms, systems, or methods.

[9]Participatory design (PD) is the overarching research tradition that emphasizes democratic involvement and shared power throughout the design process, whereas co-design refers to the specific, hands-on design activities or sessions used within (and sometimes outside of) PD to collaboratively generate ideas, artifacts, and solutions with participants.

a work with user study is primarily empirical rather than artifact-based and what constitutes a methodological contribution) and how we applied modality and visualization taxonomies. After reaching shared understanding, authors updated initial codes as needed and then independently coded the remaining corpus. We later met synchronously to review all codes, ensure consistency, and define broader categories.

Next, two authors qualitatively reviewed 45 papers, rotating roles as lead reviewer and note-taker, briefly summarizing each paper and its contributions, discussing insights, and recording emerging themes. They then reviewed and took notes on the remaining papers independently and held several follow-up meetings. Finally, the full team met to outline findings and open questions. During this process, the codes helped guide emerging themes while insights were ultimately derived by reading the papers. For instance, our codes for target communities identified education, professionals and public health communication as target audiences, leading to more in-depth analysis during this phase.

4 Descriptive Statistics of the Corpus

The following provides an overview description of our study corpus. The full corpus can be searched and reviewed through the anonymized interactive interface at https://accessvizcorpus.pages.dev/.

4.1 Publication Types, Venues & Contribution Types, and Total Publications by Year

Our dataset comprised 280 publications, including (N=179, 63.93 %) full-length papers and (N=101, 36.07 %) short papers. We analyze the year and venue of publications in figure 2 (**RQ3**). In terms of temporal distribution, (N=24, 8.57 %) papers were published between 1996 and 2005, (N=53, 18.93 %) appeared between 2006 and 2015, (N=47, 16.79 %) between 2016 and 2020, and (N=156, 55.71 %) were published from 2021 through March, 2026. The most frequent accessibility and HCI venues were ASSETS (N=53, 18.93 %), CHI (N=46, 16.43 %), and W4A (N=9, 3.21 %), while the most common visualization venues were TVCG (N=15, 5.36 %), CGF (N=11, 3.93 %), and VIS (N=6, 2.14 %).

2021 & Beyond . In 2021 and beyond, we note a significant finding that **accessible research appears to be growing exponentially** within our target research venues. Additionally, our corpus saw more work published in CHI and TVCG during this period (28 and 13 articles, respectively) than prior to 2021 (18 and 2 articles, respectively).

We also analyzed the types of research conducted, and what types of contributions were claimed. The top two types of primary research contributions were empirical (N=135, 48.21 %) and artifact (N=103, 36.79 %). Other contributions included survey (N=17, 6.07 %), theoretical (N=14, 5 %), opinion (N=8, 2.86 %), dataset (N=2, 0.71 %), and methodological (N=1, 0.36 %). Similarly, the top two types of secondary research contributions were artifact (N=73, 26.07 %) and empirical (N=72, 25.71 %), followed by theoretical (N=16, 5.71 %), methodological (N=13, 4.64 %), survey (N=6, 2.14 %), dataset (N=4, 1.43 %), and opinion (N=3, 1.07 %). Most publications centered on reading support tools (N=200, 71.43 %), with fewer—but increasing—publications focused on tools for creating accessible visualizations (N=43, 15.36 %) or on both reading and creation (N=37, 13.21 %).

2021 & Beyond . Recent years saw a higher proportion of Empirical studies as the primary contribution (55%, compared to 38% pre-2021) and a lower proportion of primary artifact contributions (26%, compared to 49% pre-2021). We found no noteworthy difference in the proportion of work focusing on read vs write pre- and post-2021.

4.2 Users and Evaluations

We identified the target communities, user engagement, and evaluation methods in our corpus and examined global trends as well as recent trends (**RQ1**).

In our corpus, most publications focused on the BLV community (N=208, 74.29 %). Additionally, there were (N=10, 3.57 %) publications that focused on screen-reader users, which included people who are BLV. Other communities included people with CVD (N=7, 2.5 %), IDD (N=5, 1.79 %), ADHD (N=2, 0.71 %), Down syndrome (N=1, 0.36 %), Autism (N=1, 0.36 %), language disabilities (N=1, 0.36 %) and disabilities more broadly (N=6, 2.14 %). Some papers also focused on creators, researchers, and visualization authors primarily (N=34, 12.14 %).

2021 & Beyond . Notably, only from 2021 onward did work centered on communities beyond visual disabilities begin to appear. Additionally, recent years saw an increase in work supporting creators in authoring accessible visualizations (7 publications post-2021, 1 pre-2021).

Figure 3 illustrates methods employed, compared between pre- and post-2021. We analyzed all methods used across the publications, with some publications employing multiple methods. The most common research methods were user studies (N=62, 22.14 %), followed by interviews (N=58, 20.71 %) and experiments (N=48, 17.14 %). Co-design (N=11, 3.93 %) and participatory design (N=6, 2.14 %) were less frequently used. Only (N=2, 0.71 %) studies employed longitudinal designs. Among (N=222, 79.29 %) papers that reported at least one method, (N=99, 35.36 %) employed two or more. Across the full corpus, (N=174, 62.14 %) papers included participants from the target audience. Studies ranged in scale from small pilots with a single participant to large-scale crowdsourcing studies with up to 432 participants (median = 12).

Age demographics were reported in (N=134, 47.86 %) studies, covering children, university students, and adults, while a substantial number of papers with user studies did not provide explicit age information (N=68, 24.29 %). Gender was explicitly reported in (N=110, 39.29 %) studies, whereas (N=92, 32.86 %) works with user studies either omitted gender or provided only partial details (e.g., "balanced" or "diverse"). Visualization experience was discussed in most publications with studies (N=130, 46.43 %), but (N=56, 20 %) did not specify it. Regarding recruitment sources, most participants were recruited from universities (N=65, 23.21 %), followed by BLV organizations and schools (N=48, 17.14 %), and mailing lists (N=33, 11.79 %).

2021 & Beyond . Notably, from 2021 and onwards, we saw a substantial increase in interview studies, user studies, surveys,

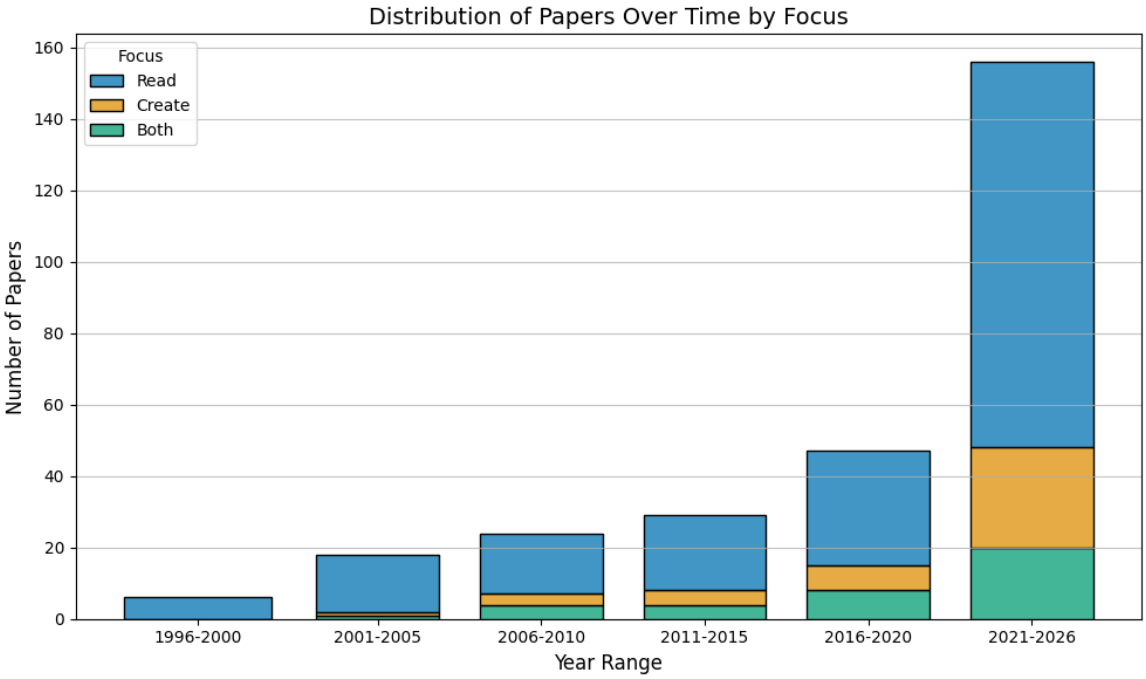


Figure 2: This stacked bar chart shows how publications from 1996 to 2026 are distributed across three focus areas: Read (blue), Create (orange), and Both (green). Reading-focused work dominates across all periods, but starting around 2016 there is clear growth in papers on creating accessible visualizations. The sharp increase during 2021–2026 reflects accelerated activity across all three areas. In our corpus, the average number of papers increased from 9.4 in 2016–2020 to 29.2 in 2021–2025 (210.6% increase). For context, averaged over 2016–2020, CHI accepted 659 papers, increasing to 914 over 2021–2025 (38.7% increase). Similarly, ASSETS increased from 33 to 55 (66.7% increase), while VIS remained relatively stable, increasing from 122 to 123 (0.8% increase).

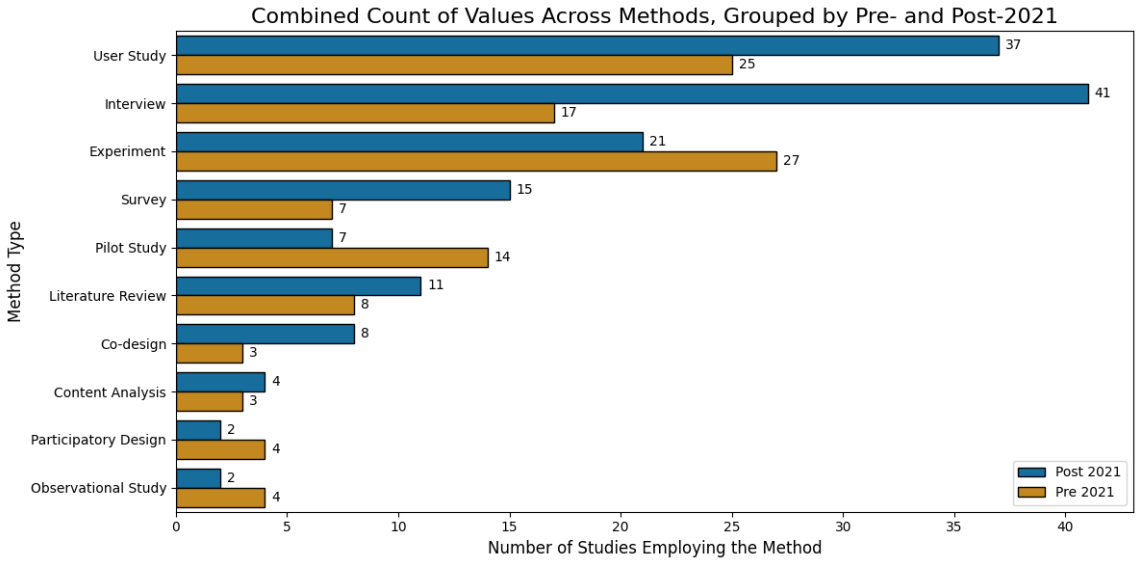


Figure 3: Horizontal bar chart showing more common methods, comparing pre- and post-2021, including user studies, interviews, and experiments as the three most common research methods. Less frequently used approaches included co-design and observational studies.

and co-design. Additionally, during this period, our corpus saw increased engagement with BLV schools (13 post-2021, 3 pre-2021).

4.3 Visualization Types, Interaction Modalities, Devices, and Use of AI

Across our corpus we identified the types of visualizations supported, interaction modalities and devices used, and whether the research employed AI (**RQ2**).

The majority of research focused on basic charts (e.g., bar, line and pie) (N=199, 71.07 %) and data visualization broadly (N=104, 37.14 %), followed by maps (N=56, 20 %), and tables (N=16, 5.71 %). Although papers that focused solely on diagrams were excluded, some works addressed diagrams in addition to other visualizations (N=48, 17.14 %).

2021 & Beyond . In recent years, a higher proportion of works have aimed to support broader data visualization (42% of our post-2021 corpus, compared to 31% pre-2021), rather than specific chart types. Only a small portion of papers (N=10, 3.57 %) aimed to support more complex, scientific visualizations.

Among specific modalities, combined speech- and text-based interactions, including text-to-speech, screen-reader, and question-answer (QA) systems, were most common (N=158, 56.43 %), followed by tactile (N=134, 47.86 %), sonification (N=67, 23.93 %), haptics (N=53, 18.93 %), and braille (N=17, 6.07 %). Nearly half of the publications focused on more than one modality in their work (N=135, 48.21 %). As illustrated in Fig. 4, there has generally been a steady increase in support across modalities.

2021 & Beyond . Recent years saw a sharp increase in speech- and text-based methods. These methods include AI-based question-answering systems which may account for some of this growth. This growth is further reflected by the increase in the proportion of works employing speech- and text-based interactions (60% post-2021, 52% pre-2021). In contrast, despite its raw growth, our post-2021 corpus saw a notable decline in the proportion of studies employing tactile and haptic modalities (35% post-2021, 65% pre-2021).

In terms of interfaces and devices, the most common interface was desktop/web (N=109, 38.93 %). (N=77, 27.5 %) studies focused partially or fully on mobile platforms and tablets. Other devices included tactile prints (N=78, 27.86 %), wearables (N=11, 3.93 %), and refreshable tactile displays (N=11, 3.93 %). Some papers also focused on specific web applications, e.g., Microsoft Excel (N=4, 1.43 %), platforms, e.g., computational notebooks (N=3, 1.07 %), and programming libraries (N=4, 1.43 %).

Given the rapidly changing landscape of AI in the past five years, we examined the use of AI in accessibility research. Our corpus contained (N=48, 17.14 %) publications that incorporated AI into the design of their tools with about two-thirds occurring after 2021 (33 post-2021, 15 pre-2021). Of these, a smaller subset of publications (N=10, 3.57 %) focused specifically on AI-based question–answering systems.

5 Findings: Communities and Use

In this section, we report on qualitative findings derived from close readings of the 280 papers in our corpus. We categorize our findings in accordance with our first two research questions. Broadly, we identify changes over the last five years in the **human experience** (**RQ1**) and the **technologies** (**RQ2**) being used and studied. Table 2 summarizes the findings.

5.1 Human Experience

We found many changes in accessibility research in the data visualization community that affect both the people involved in the studies and those served by the goals of the research (**RQ1**). We find that the characterization of user needs, the methods of engaging with users, the context of use of accessible artifacts, and the communities involved in research have all seen significant research changes in the last five years. In this section, we tag findings as relating to the user U, the research methods M, and considerations of equity in research E.

5.1.1 *Characterizing Considerations of User Needs and Engagement Practices.*

Enabling "Born Accessible" Design. E In 2023, Lazar [144] formalized two models for accessible content: the "born accessible" model and the "retrofitting" model. Born accessible design treats accessibility as a core goal from the start. Retrofitting fixes existing technology after the fact. Lazar argues that early attention to accessibility offers richer and more flexible access for more people and does so at lower cost. Chundury et al. [50] report similar findings in professional settings, noting that many visualization platforms remain inaccessible.

Prior to 2021 A substantial portion of our corpus focuses on retrofitting existing visualizations, often on the web, to enable the use of assistive technology [47, 117]. These instances are often born out of necessity - the visualizations are already published without considering accessibility, e.g., in technical documentation [178] and printed charts [47]. Under the retrofitting model, many early approaches focus on directly translating visual representations into accessible modalities. We frequently see translational approaches in tactile representations from automated charts [139, 140, 263]. Engel and Weber [72] and Xu et al. [275] note, however, that tactile graphics cannot simply replicate visual encodings and require distinct empirical design guidance.

2021 & Beyond . Recent work still often employs the retrofitting model - the need for access to existing materials has not changed. For example, Punyabukkana and Suchato [195] retrofitted an entire engineering curriculum to support a BLV student. Such solutions are likely to encounter barriers from inconsistencies in visualization design, implementation, and application of accessibility guidelines [50, 165]. Towards addressing this, we note a shift to support accessibility as a first-order design consideration. To encourage early adoption of accessibility design, several tools support rich accessibility at the authoring stage via visualization equivalence [203] and multi-modal systems [215, 295]. Better methods for auditing accessibility [6, 67] and training designers [218] have also been proposed.

Accessibility is not one-size-fits-all. E Across the corpus, accessibility emerges as not one-size-fits-all [296]. Studies note that, although there are some preferences exhibited by the group at large, people still bring diverse needs shaped by personal experience and

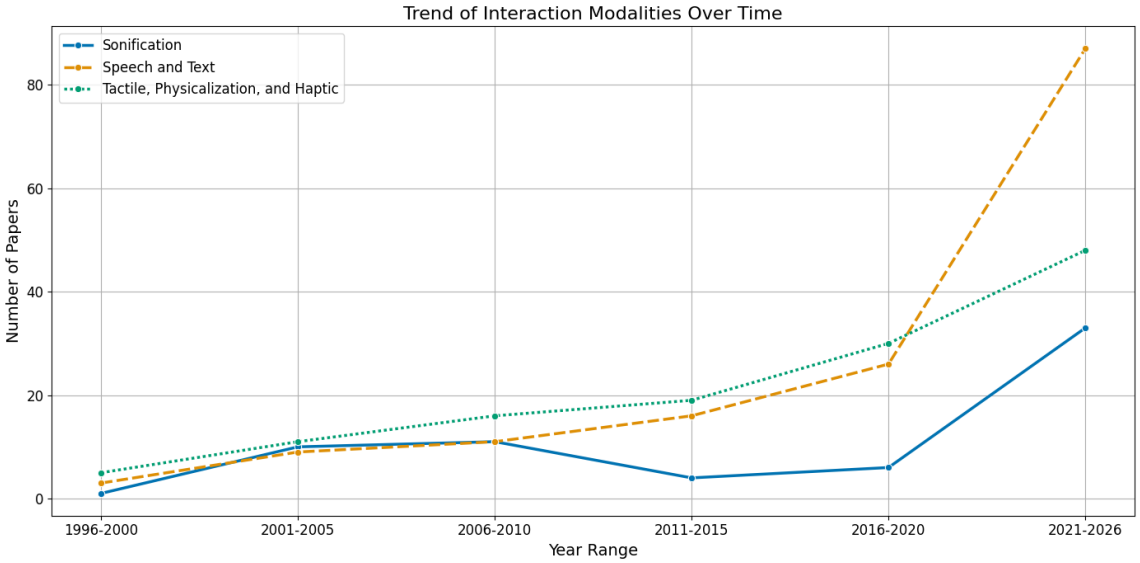


Figure 4: Line chart showing how the number of papers using sonification (blue solid line), speech and text (orange dashed line), and tactile, physicalization, and haptic (green dotted line) interaction methods changed across six time periods spanning 1996 to 2026. While sonification and tactile/haptic modalities grew steadily, speech- and text-based methods experienced a much sharper and more recent rise, particularly after 2011 and accelerating dramatically beginning in 2021.

visualization tasks. Prior to 2021 we saw instances of this in audio representations [262] and color [15, 84], and more broadly [296].

2021 & Beyond. In more recent years, several other studies identified that universal design often falls short of universal access [265], with users expressing a desire for configurations to customize data representations [223] , including sonification [221], conversational interfaces [214], personal health tracking [17], screen-reader experiences [121], among others [5, 166, 193]. Additionally, several studies found that personalization enhanced comprehension [132, 254] and that access to representations in multiple modalities helps enable personalization [215]. Recently, Elavsky et al. [69] introduced the concept of "softerware", a design approach that provides end-users with agency to customize and adapt visualizations to their needs and preferences.

When and How are Final Users (Not) Engaged? (M) As demonstrated in Fig. 3, the majority of studies engaged users through user studies, experiments, and evaluations. Few studies employed participatory or co-design. However, recent shifts in discourse around equitable research emphasize the need for user-centered and participatory design methods, equitable compensation, and alignment with disability communities' lived experiences [157, 212]. In their position paper, Angerbauer and Sedlmair [16] argue for shifting participatory design approaches from "being like" to "being with". In our corpus, seven participatory or co-design studies appeared prior to 2021 [30, 40, 83, 94, 113, 198, 296], whereas 10 were published 2021 and beyond [49, 214, 222, 232, 236, 247, 271, 279, 285, 294]. Our findings support the earlier calls for greater employment of participatory design in accessibility research [265].

5.1.2 *Context of Use.*

Pedagogy and Educational Use. (U) We found a corpus of work centered around designing and studying methods and pedagogical practices for enabling accessible data visualization in the classroom [13, 138]. [10] Our findings demonstrate a shift from primarily tactile approaches prior to 2020, to a broader range of modalities in recent years.

Prior to 2021. Early work put significant focus on enabling tactile graphics, typically with multi-modal interaction [29, 94, 143, 189]. However, early studies identified challenges for introducing tactile graphics into the classroom, such as durability [198] and formatting [90]. To address these issues, many approaches demonstrate the benefits of pairing tactile graphics with complementary audio and tactile feedback [52, 90, 198, 249] or other multi-modal representations [32, 127].

2021 & Beyond. In more recent work we still see some continued emphasis on tactile representations, in areas such as statistics [77] and computer science education [55]. Additional technologies have been studied, including a mobile audio-tactile system [172], skin-stretch haptics with audio [40], other wearable devices [240, 241], and a general push towards cheaper, adaptable alternatives to support long-term inclusion in resource-constrained educational settings [74, 195].

Beyond tactile representations, sonification has been studied in remote learning contexts [182] and for storytelling settings [105]. Recent studies also investigate pedagogy for teaching accessible visualization creation and design to students [25, 33].

[10] We recognize that our decision to focus solely on data visualization and HCI/Visualization publishers excludes a corpus of work on creating accessible diagrams in the classroom [128] and may overlook education-centric venues.

Table 2: Summary of user-centered and technological findings emerging from the qualitative analysis of papers described in Section 5, comparing work published prior to 2021 and in 2021 & beyond.

Finding	Prior to 2021	2021 & beyond
User-centered findings		
Design approach	Largely retrofitted existing data visualizations. Many systems translated visual encodings into tactile or assistive formats.	Retrofitting remains common. More work treats accessibility as a first-order design concern through authoring tools, audits, and training.
Personalization	Varied preferences for audio, color, and other representations.	Customization and multimodal access receive greater emphasis. Personalization is often found to improve comprehension.
User engagement	User studies, experiments, and evaluations dominated. Participatory and co-design methods were rare.	Participatory and co-design research increased. Equity, compensation, and lived experience receive more attention.
Educational use	Research emphasized tactile graphics, often paired with audio or haptic feedback. Durability and formatting were common challenges.	Research includes mobile audio-tactile systems, wearables, sonification, and lower-cost tools. Studies also examine teaching accessible visualization design.
Professional & public use	Little work addressed professional workflows. Accessible web visualization was a recurring goal.	Studies identify gaps in professional tools and workflows. COVID-19 visualizations exposed barriers in public information access.
User communities	Research centered on people who are BLV. Work beyond BLV communities mainly addressed CVD.	Research expands to IDD, language disabilities, ADHD, and seizure risks. More studies establish perceptual and behavioral baselines.
Technological findings		
Tactile & haptic access	Work often translated charts into tactile graphics or physicalizations. Few studies established baseline perceptual guidance.	Research increasingly establishes tactile mappings and perceptual guidance while expanding authoring support.
Sonification	Systems supported specific data types and proposed foundational design guidance.	Studies examine acoustic parameters, usability, spatial audio, audio load, and narrative design.
Speech and text	Work focused on summary generation and chart navigation. Later systems introduced richer screen-reader navigation.	Research addresses alt-text quality, text presentation, automated summaries, and richer screen-reader experiences.
Multimodal access	Audio-tactile systems were common and often outperformed individual modalities. Some systems supported multimodal authoring.	Work expands to mobile access, dynamic feedback, multimodal authoring, and support for multiple input modalities.
AI and automation	Models supported chart classification, data extraction, summary generation, and interaction with tactile charts.	AI supports authoring, automatic accessibility enhancement, rich descriptions, and conversational interfaces. Errors, bias, and security risks also emerge.
Complex visualizations	A small body of work explored coordinated views, node-link diagrams, glyphs, geospatial data, and scientific visualizations.	More work addresses uncertainty, 3D fields, heatmaps, tables, and design guidance. Overall support remains limited.

Professional Use . U Another recurrent theme in our analysis is the gap between research design and practical support for professionals in data-related positions with little work prior to 2020.

2021 & Beyond . In our corpus, this theme surfaced almost exclusively in 2021 or later. By eliciting experiences through online surveys [11, 50], analyzing accessibility of academic articles [279], considering the authoring of academic articles [173] and computational notebooks [192], recent research has surfaced gaps in the toolchain of available accessible professional software and workflows. These gaps have influenced career paths, inhibited research progress and presented barriers for accessing the work of peers [44].

Public Dissemination of Information . U Though a substantial portion of work prior to 2021 aimed to enable accessible web visualization [220, 239], the COVID-19 pandemic served as the first major test of the application of accessibility technologies and guidelines in public-facing visualizations of critical information [60, 284].

2021 & Beyond . Several post-pandemic studies found the BLV community to be particularly under-served when it came to effectively disseminating information via visualizations during this time of crisis [108, 163, 233]. For example, Fan et al. [76] studied the accessibility barriers for screen-reader users in online COVID-19 visualizations, finding that only 14% of top results were very accessible.

5.1.3 User Communities: Perception, Behavior, and Preference. To better support use and enable born accessible design, visualization designers must first understand the baseline perceptions, preferences, and behaviors of readers of varying ability.

Prior to 2021 , the vast majority of work in our corpus focused on communities with visual impairments, as similarly noted by Wimer et al. [265], and often focused on building and evaluating individual tools for accessible visualization rather than establishing baselines to guide effective design.

2021 & Beyond . From 2021 onwards, we see a significant increase in perceptual and behavioral studies aiming at setting baselines for perception and preference, across both sensory modalities and broader user communities. The following highlights some of these studies, organized by target community.

People who are Blind or have Low Vision . U 2021 & Beyond . In recent years, we saw a substantial uptick in work on perception and behavior in accessible visualization for blind or low vision users. Most studies focused on a specific modality or approach such as text-based support [132], establishing foundational guidelines for how people read or interact with visualizations in that format [135]. We leave greater discussion of modality-specific studies for Sec. 5.2.1.

As noted by Kim et al. [136], though many studies broadly target BLV communities, designs are often geared towards and evaluated by people who are blind rather than people with low vision. We found a similar trend in our corpus, with only two studies specifically targeting individuals with low vision [166, 262].

Beyond Visual Impairment . U Prior to 2021 , the only accessibility support outside of visual impairment in our corpus came from accessibility for color vision deficiencies. These studies focused on designing color vision deficient-friendly mappings - both static [208] and personalized [84, 255] as well as color differentiability under color vision deficiencies [202].

2021 & Beyond . From 2021 onwards, our corpus contained continued research for color vision deficiencies relative to the subjectivity of color accessibility [15] and the need for personalization [231]. Additionally, recent studies have broadened community support to include people with intellectual and developmental disabilities, language disabilities, ADHD, as well as people prone to seizures. Research into the perception of and accessibility needs for people with intellectual and developmental disabilities has only begun to emerge in the last few years, with studies characterizing everyday data encounters [273], studying perceptual strategies used by people with IDD [269], identifying what makes a visualization (in)accessible [12, 268, 270], and designing cognitively accessible visualizations [271, 272]. Recent studies have also examined the barriers to accessibility for people with language disabilities [267], the effect of ADHD on perception and comprehension of charts [17, 251] as well as the seizure-inducing risk of animated and interactive data visualizations [235].

5.2 Technological Changes

Recent rapid changes in both hardware and software have led to significant changes in research into accessibility for data visualization (**RQ2**). We find that the types of sensory modalities studied, the types of technologies used to create accessible representations and experiences, and the complexity of data representations being studied have all led to significant changes in recent years. We broadly categorize these changes in sensory modalities and devices to allow for greater personalization to diverse user needs P, trends in technological approaches relating to AI-driven software T, and the support for visualization and interaction beyond basic statistical charts C.

5.2.1 Non-Visual Sensory Modalities. As outlined by Kim et al. [137], there are two primary non-visual sensory modalities for conveying data: tactile and auditory. Tactile approaches leverage touch as the primary sense, such as with embossed or raised line graphics, as well as haptic feedback such as vibrations. Auditory modalities contain two types of audio: speech-based and non-speech-based audio, often called sonification. We scope speech-based audio to include any technologies that are speech-based by design or that provide text that can be read aloud by a text-to-speech device. In line with findings from Vargas and Trevisan [254], we found that the majority of studies identify multi-modal representations and systems to be more effective than singular modalities. In the following, we highlight key trends in each modality but refer readers to prior reviews, such as Kim et al. [137] for deeper descriptions.

Tactile. P Tactile and haptic visualizations are often studied in the context of supplementing traditional visualizations to foster engagement, as demonstrated in the multiple reviews of this space [34, 184, 254]. Our examination focuses on technologies specifically designed to enable access to data visualizations.

Prior to 2021 . In line with prior findings, we identify several strategies to promote active exploration rather than static reading [102, 142, 244, 282]. Beyond the findings of prior studies, we note a substantial portion of early research focused on translating visual encodings into corresponding tactile representations in both 2D charts [95, 139, 140, 263] and 3D physicalizations [29, 113], as discussed in Sec. 5.1.1. Subsequently, we note very few studies establishing baseline perceptual guidelines. Those that exist largely focus on chart-specific perception [18, 24, 228], with only one aiming to establish the effectiveness of material channels (friction, stiffness, and texture) for data representation [257]. Additionally, several works support BLV users in creating tactile graphics through various mediums including: tangible interfaces [63, 169], grammar-based tools [181], rapid 3D-printed prototyping [30, 109], automated conversion from textbook figures [117] and text files [93], refreshable tactile displays [156], and textured tokens [75].

2021 & Beyond . Recent years saw a shift from specific instances of tactile approaches, to baseline design guidance and broad authoring support. Several recent studies aim to better understand tactile perception of graphics [21, 118, 164, 290] and outline effective mappings from data to tactile channels [48, 65]. These studies signify a shift from the traditional "translational" approach of prior works towards more "born-accessible" design by establishing the effectiveness of low-level tactile channels, as well as strategies for perception. To that end, we also note advances in authoring tactile visualizations. Pittarello and Semenzato [191] designed a physicalization toolkit to build and explore data representations from tabular descriptions, offering design guidelines for creating accessible, modular physicalizations [190]. Other works provide tools to more easily create graphics with tactile feedback during the creation process [229] or through more logical, automated means [42, 129].

Auditory - Non-speech. P

Non-speech audio or sonification aims at leveraging non-speech channels for representing data and conveying insights [70], often coupled with other modalities to drive interaction [254].

Prior to 2021. In earlier works, sonification was used to enable exploration of particular types of data, such as 3D data [174] or geo-referenced data [287]. Various works proposed foundational guidance for the design, use, and translation of sonified data [86, 134, 289], including specific visualization paradigms such as multiple coordinated views [288] or tours through the data [87].

2021 & Beyond. In recent years, we see an expansion of foundational perceptual and behavioral studies to define the design space of effective sonifications in terms of acoustic parameters [256, 259, 291] and usability [219, 221]. Further studies consider environmental effects, including reducing audio load via importance-based audio emphasis [31], leveraging spatial audio to improve trend identification [206], and the impact of musical background on sonification perception [111]. Kim et al. [133] distill the findings of prior perceptual studies into Erie, a grammar for authoring effective and expressive data sonifications. Other recent work explores guidance [110] and design principles in sonification narratives [232].

Auditory - Speech. (P) While speech and text descriptions of visualizations are one of the most frequently supported accessibility paradigms, we note that Kim et al. [136] find that existing visualization tools rarely support accessibility beyond alt text.

Prior to 2021. Earlier studies in our corpus largely focus on strategies for effective summary generation [107, 175, 246] and navigation of charts [79, 80].

In the late 2010's, some work shifted from text generation to screen-reader navigation, aiming to enable hierarchical navigation of visualizations [83] and interfaces for rich interaction [296] on statistical charts.

2021 & Beyond. We note a substantial broadening of the landscape of research around speech-based audio in recent years.

Recent studies note issues with the adoption and execution of alt-text in the wild, such as low prevalence and inconsistent quality of alt text for figures in CHI publications [264], insufficient detail for rich SVG visualizations [238], and a focus on explaining visual elements rather than underlying data [123]. Building on these limitations, we note subsequent research in three primary directions: (1) effective presentation of text descriptions [149, 158, 194, 234], (2) (automatic) summary generation [171, 230, 274], and (3) enabling richer screen-reader experiences [96, 155, 224, 247, 294].

Multi-modal access. (P) Kim et al. identified that approaches leveraging multiple sensory modalities often provide richer and more interactive experiences [137]. In our analysis, we found similar trends, with repeated findings that multi-modal approaches excelled over individual modalities.

Prior to 2021. In earlier research, the most common pairing we saw was tactile graphics and navigation with audio feedback, such as non-speech auditory cuing to guide navigation [53, 145, 286], speech audio to provide detail to tactile interactions [90, 103, 104, 162], or a combination of both [3, 292]. Finally, several works paired tactile graphics with both audio and haptic feedback [1, 2, 125, 130] with some finding that multi-modal approaches better support both active exploratory interaction and passive guided interaction [2]. Many of these studies found multi-modal tactile-audio interaction to be more effective than individual modalities, leading to higher accuracy and faster task performance [99, 116, 280, 281]. Finally, two studies in our corpus support BLV users in authoring of audio-tactile charts [71, 150].

2021 & Beyond. Recent studies still trend towards multi-modal audio-tactile approaches [61, 160, 207], though some integrate more recent technological advances enabling fingertip recognition [199] or dynamic feedback [200, 201]. Two emerging areas we saw in recent years were multi-modal access to graphs on mobile devices [49, 196, 285] and authoring tools for creating multi-modal data representations [215, 222, 225, 295]. Finally, rather than providing multi-modal outputs, Elavsky et al. [68] examine how to support interaction from a variety of input modalities. These studies highlight the importance of multi-modal access to visualizations to enable personalization to the skills and preferences of individual users.

5.2.2 *Automated and AI-enabled Accessibility.*

Intelligent Technologies and Software. (T) The drastic changes in AI- and data-science-driven tools have resulted in new assistive technologies incorporating automation. Our examination focuses on automated HCI studies with accessibility at the forefront.

Prior to 2021. Studies incorporating predictive models typically focused on generating datasets and machine learning artifacts (e.g., chart classifiers and data extractors) for downstream access support [41, 46, 131], intelligently decomposing visualizations to generate natural language summaries [38, 92, 100, 101], and applying machine vision to track user navigation [90] and enable gestural interaction [227] in tactile charts.

2021 & Beyond. In contrast, recent years saw a wide expansion of AI-enabled accessibility, expanding beyond summary generation to support more efficient authoring of accessible visualizations, automatic enhancement of accessibility, and rich natural language interactions for comprehending visualizations. To lower the overhead of designing accessible visualization, some studies automated components of authoring [27, 146]. The second direction identified was automatic enhancement of accessibility in existing visualizations that lack accessibility features. Several works aimed at automatically generating rich text descriptions [8, 62, 81, 180]. Generative models have also been used to transform non-text data such as tables [261] or generate tags or semantic representations of images [176, 187]. Other contributions include new datasets to support generation of alt-text [39, 45]. The third direction we identified was the use of LLM-driven conversational interfaces to enable rich natural language interactions [97, 98, 201, 214].

There has been increased awareness of the limitations of AI-generated content which may limit the value of this thrust of research. In studies of AI-generated alt-text, poor classification of graphical features has been found to lead to incorrect and misleading alt-text [62, 230]. In addition, recent research has uncovered implicit societal bias against disabled communities in large language models that could impact the usefulness and potential harm of generated descriptions [91]. In general, the incorporation of potentially transformational capabilities of AI holds much promise, but could potentially exacerbate equality issues by providing substandard solutions with unanticipated consequences [9, 66]. For example, BLV users of visual assistance technology may be exposed

to greater security risks [7]. Potential harms should be considered in the application of AI to intelligent technologies and software.

5.2.3 *Beyond Statistical Charts and Simple Navigation.*

Expanding Accessible Chart Types. C In line with the findings of past surveys, the vast majority of approaches focus on simple charts, such as bar, line, or pie charts, and basic navigation interactions for reading static charts [137]. One of the greatest gaps we surfaced is the support for scientific visualization. Scientific visualization encompasses some of the most complex designs but our corpus revealed very few dedicated accessibility studies.

Prior to 2021 . Earlier studies did explore complex charts, including multiple coordinated views [287, 288], node-link representations [22, 52], glyphs [204, 278], and geospatial data [145, 216]. For scientific visualization, there were studies that explored color-accessible transfer function design [43], and haptic visualizations for 1D, 2D, and 3D scientific data [89].

2021 & Beyond , we observe a greater volume of work supporting more complex visualizations and interactions. Towards supporting scientific visualization, recent studies investigated uncertainty [226], accessible representations of 3D scalar fields [126], complex visualizations such as cluster heatmaps [106], complex graphs and navigation tasks via audio-tactile representations of node-link diagrams [292], and tactile representations of node-link diagrams and adjacency matrices [277]. We also saw advances in supporting accessible tables, which are often a default representation for assistive technologies such as screen-readers [120, 136, 260, 261]. Finally, we saw foundational studies to establish design guidelines for complex visualizations [233, 236].

6 Discussion

Our findings and analysis demonstrate that significant growth in accessibility research for visualizations is leading to changes in user-centered research and technologies.

Our first three research questions focused on recent changes in participant and target-population engagement **(RQ1)**, technologies used in accessible visualization **(RQ2)**, and publication volume within the visualization research community **(RQ3)**.

To answer **RQ1**, we examined changes in the *target users* considered in accessible visualization research, the *methodological* approaches used to understand these users, and the growing emphasis on *equity* as a priority for responsible and effective research. We found most earlier work focuses on people who are BLV or who use screen readers, but that some more recent work addresses creators and visualization authors, along with communities such as people with CVD, IDD, ADHD, or Down syndrome. Empirical and artifact contributions dominate the corpus, with experiments, user studies, interviews, and pilot studies as the primary methodological approaches in earlier works, and only a small number of studies use longitudinal methods. More recently, co-design and participatory design have become methods for greater involvement and equity with affected communities.

To answer **RQ2**, we examined recent changes in *AI*, *new hardware*, and the growing *variety* and *personalization* of accessible data representations. Recent progress in *AI* has caused upheaval in many computing research domains, and many new avenues of interaction have become available to researchers. *New hardware* including tactile, auditory, and haptic interfaces could enable greater *variety* and *personalization* of representations of data. We find that across our whole corpus, charts are the most common visualization type, followed by maps and tables. Text and screen-reader support remain the dominant interaction modalities and have expanded rapidly with recent AI systems, compared with sonification, haptics, tactile interfaces, and braille. The web is the most frequently studied platform, with additional work on mobile and tablet interfaces, wearables, refreshable tactile displays, and software ecosystems, e.g., Excel and notebooks.

For **RQ3**, we considered whether greater awareness of accessibility research and methods is leading to a greater volume of work contributing to the accessibility of data visualizations. We found that the rate of published works has grown exponentially within our venues of study, while those venues have only grown linearly or stayed about the same size. This suggests that accessibility will become a more significant pillar of those research venues.

Based on trends and identified gaps in recent research, we present open questions to help drive and organize future research in this space (**RQ4**). We present open questions and future research directions organized, consistent with our qualitative findings, into user-centered and technological themes.

6.1 Users and Research Methods

Our findings underscore a growing need to move away from visualization-first representations and toward born-accessible design [144]. In line with prior work, our findings also reinforce that personalization, participatory design, and broader community inclusion [265] are essential for research on accessible data visualization. Finally, although work for BLV readers remains essential, expanding attention to cognitive disabilities, ADHD, CVD, and other groups is needed to build more inclusive design principles reflective of the diverse barriers people encounter in everyday data use [265].

By focusing on usage contexts, our analysis surfaced insights for both pedagogical and professional settings.

We also highlight a persistent gap between research innovations and the needs of professionals and the general public. Many tools used in data-centric workplaces remain inaccessible, limiting participation and career trajectories for data-oriented professionals with disabilities [44], and public-facing visualizations struggle to support accessible insight generation despite nominal compliance with guidelines [108].

We outline some open questions below:

U *How do access challenges and strategies differ across disability types as more heterogeneous populations are considered?* Our research built on findings from prior surveys that noted that most accessible visualization research has centered on BLV users, with limited attention to other disability groups or intersections. More recent work included other considerations such as congenital versus acquired conditions and multi-disability contexts, suggesting that a paradigmatic shift could be needed to balance or personalize needs. For example, users with congenital blindness may develop distinct tactile exploration strategies compared to those with acquired vision loss

[213], while DeafBlind users require solutions that combine or extend modalities [56].

- (U) *How should accessible visualization tools differ for distinct contexts of use or user groups?* Example contexts of use include pedagogical or professional settings, and example user groups include children, graduate students working with scientific visualizations, professionals managing cognitive load, and older adults. While prior work highlights user diversity, systematic comparisons remain limited. Our review identified that some usage contexts such as pedagogy continue to be actively studied, but more recently, professional settings have been highlighted as a setting with a dearth of available resources. Learning frameworks such as comparative organizers, schema theory, and the distinction between declarative and procedural knowledge offer structured steps for introducing new concepts [26].
- (M) *What new evaluation frameworks can capture accessibility outcomes that unfold beyond momentary task performance?* As the context of use and user groups differ, the barriers to use might be measured by different objectives, such as fatigue or long-term learning. These outcomes should also be measured in authentic, everyday contexts. Controlled studies rarely reveal issues that emerge over sustained use, such as durability, shifting user goals, or when AI-generated summaries are "good enough" for trend scanning yet insufficient for scientific or professional precision.
- (E) *Who bears the labor of accessibility: creators, platforms, or auditors?* The development of effective and usable technology for accessible data representations should include sufficient resources and incentives for all stakeholders, including designers and creators. Prior work emphasizes embedding accessibility features into authoring tools so creators can design inclusively without relying solely on external audits [215]. Yet much of the burden still falls on people with disabilities, who must identify gaps and advocate for fixes. Open questions remain around how platforms can integrate accessibility by default and how incentives can encourage authors, toolmakers, and institutions to prioritize inclusive design.

6.2 Chart Types, Interaction Modalities, Technologies

To support the shift away from visualization-first representations, we note a complementary trend in studies defining the design space of non-visual representations, relative to perceptual and behavioral baselines. We saw this across sensory modalities, but more heavily in tactile and speech-based audio, where existing approaches were often tied to prior visual encodings.

Subsequently, we saw an increase in authoring support for customizable, multi-modal visualizations. As noted in Sec. 5.2.1, multimodal representations are frequently found to be more effective and engaging than individual sensory modalities.

AI presents an opportunity for enriched accessibility along several axes, such as automated integration of accessibility features into existing visualization [62], semi-automatic or guided authoring of accessible visualizations [27], and enriched interactions for navigating and querying visualizations [97].

Finally, we note a continued lack of support for complex visualizations and rich data interactions. Surprisingly little work even mentions scientific visualizations [39, 45, 126], despite their importance in many educational and professional domains.

We outline some specific questions below:

- (C) *How can accessibility techniques support complex visualizations?* Prior work notes the need to move beyond simple chart types [50], and some studies show that people with disabilities may prefer familiar or lower-load visualizations [272]. Yet our review found only a small amount of work on complex visualizations, such as scientific visualizations.
- (T) *What is the role of automation in accessible visualizations, and is it welcome?* Significant recent work has focused on automating the generation of accessible artifacts, or even automating evaluation or auditing of accessible representations [67]. At the same time, there may be significant reluctance to remove the human from the loop in the design and evaluation of these artifacts. A research challenge is how to design hybrid auditing pipelines that combine fast, automated checks with structured points for human evaluation.
- (T) *What is the design space of interactions between user and AI that prioritizes accessibility?* AI-generated chart summaries and question–answering are a popular area of research, but as they become more ubiquitous in accessibility, there are opportunities to incorporate collaborations between user and AI. Certain question-answering strategies (e.g., compare, contrast, summarize, what-if) might effectively support non-visual exploratory analysis while giving users fine-grained control over automated outputs.
- (T) *What are the limitations of material properties of tactile charts, and can they change?* We found that more chart types are being explored, and we suggest that accessible representations of complex data will be tackled in the near future. As the tactile complexity of these representations increases, material properties (e.g., friction, stiffness, sinusoidal texture) affect tactile chart readability and recognition accuracy. Prior research has shown that tactile perception is sensitive to variations in texture and friction, but systematic guidelines for their use in accessible visualization are still lacking [257].
- (P) *How can non-visual visualization systems learn from a user's interaction history to personalize and adapt?* Prior work notes the need for within-modality personalization [254], yet it remains unclear how systems—especially those powered by emerging LLM-based tools—should adapt to users' navigation patterns, task goals, and preferred levels of detail. Frameworks such as Multimodal Complementarity [258], the Information-Seeking Mantra [54], and Levels of Description [158] offer guidance for distributing information and structuring overviews and details. A challenge is enabling fluid shifts between different data needs, e.g., between fine-grained, datum-level reading and higher-level skimming.
- (P) *What fidelity is lost or preserved when exporting charts between modalities?* Our review found that translation of visual

representations to accessible representations (e.g., SVG → tactile, chart → CSV/audio) is still a common solution that is likely to continue. A persistent challenge in accessible visualization is maintaining semantic fidelity during conversion. Prior work has shown that critical information, e.g., spatial layout, perceptual grouping, or higher level statistical properties may be lost when a chart is flattened into a tactile or auditory form [263]. More research is needed to establish templates and metadata standards that can preserve perceptual levels (e.g., construction details, statistical properties) across modalities.

7 Limitations and Future Work

We limited our search to ACM, IEEE, Eurographics, Springer Nature Link, and Taylor & Francis venues because they align with our expertise, reflect our initial data-visualization focus, and offer consistent indexing and metadata standards. We acknowledge that relevant work also appears in other digital libraries, disability-focused journals (e.g., the Journal of Visual Impairment and Blindness and the Journal of Blindness Innovation and Research), and adjacent fields such as health, education, vision science, psychology, and communication studies. As a result, our review reflects primarily *intradisciplinary* HCI trends, and future research could take a more *interdisciplinary* approach.

Our methodology relied on qualitative coding and descriptive statistics, which allowed deeper engagement with individual contributions but carries risks of subjectivity and missing relevant work. Future reviews could incorporate quantitative techniques such as bibliometrics or topic modeling. In line with recent surveys addressing similar research questions [10, 159], we did not fully adopt standardized systematic review reporting frameworks such as PRISMA [245] or QUOROM [51], which are widely used in health-related reviews. However, we incorporated the relevant elements for scoping reviews identified by prior work [252]. Including a full PRISMA protocol in future multidisciplinary surveys may better align HCI reviews with broader reporting standards and make them more accessible to clinical, educational, and policy communities. Further, although reviews involving multiple coders may use inter-rater reliability (IRR) to assess coding consistency, we did not calculate IRR because our codes were descriptive rather than interpretive [167]. Instead, we discussed coding decisions during regular meetings and resolved disagreements through consensus.

Our inclusion and exclusion criteria may have led to the omission of borderline papers. Foundational visualization work not explicitly framed around accessibility but potentially insightful, such as abstract graph-drawing models or advances in sonification, also fell outside our scope. Because our review covers work only up to March 2026, and accessible visualization is evolving quickly with advances in AI, generative models, and new tactile hardware, regular updates will be needed to capture emerging developments.

Our coding and data collection did not include an accounting of disability justice; for example, HCI research that considers oppression on a single identity axis (e.g. race-only, gender-only, disability-only) can propagate harmful biases rather than serve the intended user base [242, 243]. This can result in research that produces "disability dongles", a term coined by Liz Jackson to describe artifacts that do not address real needs of disabled people but glorify their creators through quick fixes [115]. There are many dimensions of disability justice beyond axes of identity that would be important to consider, including appropriate compensation and citational justice [293], factors affecting practicality of design [23], and downstream effects of technology on climate change [73]. By not engaging with these issues in our analysis, we risk propagating biases and injustices in our findings. We acknowledge this as a limitation, but did not include these dimensions in our coding process because the scale of our included set of papers did not allow for a deeper study of each paper to identify factors that may be implicit or inconsistently reported.

8 Conclusion

We present a survey of 280 HCI, data visualization, and accessibility publications from ACM, IEEE, Eurographics, Springer Link, and Taylor & Francis venues that focus on HCI contributions to accessible data visualization. This review highlights a breadth of work, with a strong emphasis on empirical and artifact contributions but limited engagement in participatory design. Most research in this space focuses on BLV communities and simple visualizations, with recent attention to people with non-vision-based disabilities but limited support for complex visual interactions and scientific visualization. We distill our findings into a set of open questions for future research. Topics include integrating AI-driven descriptions while preserving fidelity across modalities, personalization across and within modalities, and better visualization auditing tools.

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